Homework 3

Due Date: 2:00 pm, May 18 (Monday)

Problem 1. For each of the following linear programming models, give your recommendation on which is the more efficient way, probably, to obtain an optimal solution: by applying the simplex method directly to this primal problem or by applying the simplex method directly to the dual problem instead. Explain.

(a) Maximize

$$Z = 8x_1 - 3x_2 + 5x_3,$$

subject to

$$\begin{aligned} 2x_1 - x_2 + 3x_3 &\leq 28, \\ x_1 + 2x_2 + x_3 &\leq 34, \\ 4x_1 + x_2 + 2x_3 &\leq 50, \\ 3x_1 - 2x_2 + x_3 &\leq 22, \\ x_1 + 3x_2 + 2x_3 &\leq 45, \\ 5x_1 + 2x_2 + x_3 &\leq 60, \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

(b) Maximize

$$Z = 3x_1 + 6x_2 + 2x_3 + 5x_4 + 4x_5 + 7x_6,$$

subject to

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + x_4 + 4x_5 + 2x_6 &\leq 18, \\ 3x_1 + x_2 + 2x_3 + 5x_4 + x_5 + 4x_6 &\leq 24, \end{aligned}$$

and

$$x_j \geq 0, \qquad \text{for } j = 1, 2, \ldots, 6.$$

Solution

For a primal linear programming problem in the standard maximization form

$$\max\ c^T x,$$

subject to

$$Ax \leq b, \qquad x \geq 0,$$

the corresponding dual problem is

$$\min\ b^T y,$$

subject to

$$A^T y \geq c, \qquad y \geq 0.$$

Thus, if the primal problem has m functional constraints and n decision variables, then its dual problem has n functional constraints and m decision variables. Since the simplex method is generally more efficient when there are fewer functional constraints, we compare the number of constraints in the primal and dual problems.

Summary.

Part	Primal Size	Dual Size	Recommendation
(a)	6 constraints, 3 variables	3 constraints, 6 variables	Solve the dual
(b)	2 constraints, 6 variables	6 constraints, 2 variables	Solve the primal

Problem 2. Consider the following problem.

$$\text{Maximize} \quad Z = -2x_1 - x_2 - 3x_3,$$

subject to

$$\begin{aligned} 2x_1 + x_2 + x_3 &\leq 15, \\ x_1 + 3x_2 - 2x_3 &\leq 4, \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

(a) Construct the dual problem.

(b) Use duality theory to show that the optimal solution for the primal problem has $Z \leq 0$.

Solution

The primal problem is in the standard maximization form

$$\max\ c^T x,$$

subject to

$$Ax \leq b, \qquad x \geq 0.$$

Therefore, its dual problem is

$$\min\ b^T y,$$

subject to

$$A^T y \geq c, \qquad y \geq 0.$$

For this problem,

$$c = \begin{bmatrix} -2 \\ -1 \\ -3 \end{bmatrix}, \qquad b = \begin{bmatrix} 15 \\ 4 \end{bmatrix},$$

and

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & -2 \end{bmatrix}.$$

(a)

The dual problem is

$$\text{Minimize} \quad W = 15y_1 + 4y_2,$$

subject to

$$\begin{aligned} 2y_1 + y_2 &\geq -2, \\ y_1 + 3y_2 &\geq -1, \\ y_1 - 2y_2 &\geq -3, \end{aligned}$$

and

$$y_1 \geq 0, \qquad y_2 \geq 0.$$

(b)

We show that the optimal objective value of the primal problem satisfies

$$Z \leq 0.$$

Consider the dual solution

$$y_1 = 0, \qquad y_2 = 0.$$

This solution is feasible for the dual problem, because

$$\begin{aligned} 2(0) + 0 = 0 &\geq -2, \\ 0 + 3(0) = 0 &\geq -1, \\ 0 - 2(0) = 0 &\geq -3. \end{aligned}$$

The corresponding dual objective value is

$$W = 15(0) + 4(0) = 0.$$

By weak duality, for any feasible solution of the primal problem and any feasible solution of the dual problem,

$$Z \leq W.$$

Since the dual problem has a feasible solution with

$$W = 0,$$

it follows that every feasible solution of the primal problem satisfies

$$Z \leq 0.$$

Therefore, the optimal objective value of the primal problem satisfies

$$Z^* \leq 0.$$

Problem 3. Consider the following problem.

$$\text{Maximize} \quad Z = 4x_1 - 12x_2,$$

subject to

$$x_1 - 3x_2 \leq 2,$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

(a) Construct the dual problem, and then find its optimal solution by inspection.

(b) Use the complementary slackness property and the optimal solution for the dual problem to find the optimal solution for the primal problem.

(c) Suppose that c_1, the coefficient of x_1 in the primal objective function, actually can have any value in the model. For what values of c_1 does the dual problem have no feasible solutions? For these values, what does duality theory then imply about the primal problem?

Solution

The primal problem is

$$\max\ 4x_1 - 12x_2,$$

subject to

$$x_1 - 3x_2 \le 2,$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0.$$

This is in the standard maximization form

$$\max\ c^T x,$$

subject to

$$Ax \le b, \qquad x \ge 0.$$

Therefore, its dual problem is

$$\min\ b^T y,$$

subject to

$$A^T y \ge c, \qquad y \ge 0.$$

Here,

$$A = \begin{bmatrix} 1 & -3 \end{bmatrix}, \qquad b = 2, \qquad c = \begin{bmatrix} 4 \\ -12 \end{bmatrix}.$$

(a)

Since there is only one primal functional constraint, the dual problem has one dual variable, denoted by y.

The dual problem is

$$\text{Minimize} \quad W = 2y,$$

subject to

$$y \ge 4,$$
$$-3y \ge -12,$$

and

$$y \ge 0.$$

The second constraint can be rewritten as

$$y \leq 4.$$

Hence the dual constraints are

$$y \geq 4, \qquad y \leq 4, \qquad y \geq 0.$$

Therefore, the only feasible solution is

$$y^* = 4.$$

The corresponding dual objective value is

$$W^* = 2(4) = 8.$$

Thus, the optimal solution of the dual problem is

$$y^* = 4$$

with optimal objective value

$$W^* = 8.$$

(b)

We now use complementary slackness to find the optimal solution of the primal problem.

The primal constraint is

$$x_1 - 3x_2 \leq 2.$$

The corresponding dual variable is y. Since

$$y^* = 4 > 0,$$

the complementary slackness condition implies that the corresponding primal constraint must be binding:

$$2 - (x_1 - 3x_2) = 0.$$

Therefore,

$$x_1 - 3x_2 = 2.$$

The two dual constraints are

$$y \geq 4,$$

and

$$-3y \geq -12.$$

The corresponding complementary slackness conditions are

$$(y - 4)x_1 = 0,$$

and

$$(-3y + 12)x_2 = 0.$$

Substituting $y^* = 4$ gives

$$(4 - 4)x_1 = 0,$$

and

$$(-3 \cdot 4 + 12)x_2 = 0.$$

Thus,

$$0 \cdot x_1 = 0, \qquad 0 \cdot x_2 = 0.$$

These conditions impose no additional restrictions on x_1 and x_2.

Therefore, the optimal primal solutions are all nonnegative solutions satisfying

$$x_1 - 3x_2 = 2.$$

Let

$$x_2 = t, \qquad t \geq 0.$$

Then

$$x_1 = 2 + 3t.$$

Hence the set of optimal solutions of the primal problem is

$$(x_1, x_2) = (2 + 3t, t), \qquad t \geq 0.$$

For any such solution,

$$Z = 4(2 + 3t) - 12t = 8 + 12t - 12t = 8.$$

Therefore, the optimal objective value is

$$Z^* = 8.$$

Problem 4. Consider the following linear programming problem.

$$\begin{aligned}
\text{Maximize} \quad & Z = 3x_1 - 2x_2 + x_3, \\
\text{subject to} \quad & x_1 + x_2 - x_3 = 6, \\
& 2x_1 - x_2 + 3x_3 \leq 10, \\
& -x_1 + 4x_2 + x_3 \geq -2, \\
\text{and} \quad & x_1 \text{ is unrestricted in sign}, \\
& x_2 \geq 0, \\
& x_3 \leq 0.
\end{aligned}$$

(a) Use the dual construction rules directly to construct the dual problem.

(b) Convert the primal problem to standard maximization form, construct the corresponding dual problem, and show that this dual problem is equivalent to the one obtained in part (a).

(c) Verify that

$$\bar{x} = \left(\frac{26}{5}, \frac{4}{5}, 0\right)$$

and

$$\bar{y} = (2, 0, -1)$$

are optimal solutions for the primal and dual problems, respectively.

Solution

(a)

The primal problem is a maximization problem. It has three functional constraints:

$$x_1 + x_2 - x_3 = 6,$$
$$2x_1 - x_2 + 3x_3 \le 10,$$
$$-x_1 + 4x_2 + x_3 \ge -2.$$

Let the corresponding dual variables be

$$y_1, \quad y_2, \quad y_3.$$

Since the first primal constraint is an equality constraint, the corresponding dual variable is unrestricted in sign:

$$y_1 \text{ is unrestricted in sign.}$$

Since the second primal constraint is a $\le$ constraint in a maximization problem, the corresponding dual variable satisfies

$$y_2 \ge 0.$$

Since the third primal constraint is a $\ge$ constraint in a maximization problem, the corresponding dual variable satisfies

$$y_3 \le 0.$$

The dual objective function is obtained from the right-hand sides of the primal constraints:

$$\text{Minimize} \quad W = 6y_1 + 10y_2 - 2y_3.$$

Now construct the dual constraints column by column.

For x_1, since x_1 is unrestricted in sign, the corresponding dual constraint is an equality:

$$y_1 + 2y_2 - y_3 = 3.$$

For x_2, since $x_2 \ge 0$, the corresponding dual constraint is

$$y_1 - y_2 + 4y_3 \ge -2.$$

For x_3, since $x_3 \leq 0$, the corresponding dual constraint is

$$-y_1 + 3y_2 + y_3 \leq 1.$$

Therefore, the dual problem is

$$\begin{aligned}
\text{Minimize} \quad & W = 6y_1 + 10y_2 - 2y_3, \\
\text{subject to} \quad & y_1 + 2y_2 - y_3 = 3, \\
& y_1 - y_2 + 4y_3 \geq -2, \\
& -y_1 + 3y_2 + y_3 \leq 1, \\
\text{and} \quad & y_1 \text{ is unrestricted in sign}, \\
& y_2 \geq 0, \\
& y_3 \leq 0.
\end{aligned}$$

(b)

We now convert the primal problem to standard maximization form.

Since x_1 is unrestricted in sign, write

$$x_1 = x_1^+ - x_1^-,$$

where

$$x_1^+ \geq 0, \qquad x_1^- \geq 0.$$

Since

$$x_3 \leq 0,$$

write

$$x_3 = -u_3,$$

where

$$u_3 \geq 0.$$

The objective function becomes

$$\begin{aligned}
Z &= 3(x_1^+ - x_1^-) - 2x_2 + (-u_3) \\
&= 3x_1^+ - 3x_1^- - 2x_2 - u_3.
\end{aligned}$$

The equality constraint

$$x_1 + x_2 - x_3 = 6$$

becomes

$$x_1^+ - x_1^- + x_2 + u_3 = 6.$$

To convert it into standard maximization form, replace it by two inequalities:

$$x_1^+ - x_1^- + x_2 + u_3 \leq 6,$$

and

$$-x_1^+ + x_1^- - x_2 - u_3 \leq -6.$$

The second primal constraint becomes

$$2x_1^+ - 2x_1^- - x_2 - 3u_3 \leq 10.$$

The third primal constraint is

$$-x_1 + 4x_2 + x_3 \geq -2.$$

Substituting $x_1 = x_1^+ - x_1^-$ and $x_3 = -u_3$, we get

$$-x_1^+ + x_1^- + 4x_2 - u_3 \geq -2.$$

Multiplying both sides by -1 gives

$$x_1^+ - x_1^- - 4x_2 + u_3 \leq 2.$$

Therefore, the standard-form primal problem is

$$\begin{aligned}
\text{Maximize} \quad & Z = 3x_1^+ - 3x_1^- - 2x_2 - u_3, \\
\text{subject to} \quad & x_1^+ - x_1^- + x_2 + u_3 \leq 6, \\
& -x_1^+ + x_1^- - x_2 - u_3 \leq -6, \\
& 2x_1^+ - 2x_1^- - x_2 - 3u_3 \leq 10, \\
& x_1^+ - x_1^- - 4x_2 + u_3 \leq 2, \\
\text{and} \quad & x_1^+, x_1^-, x_2, u_3 \geq 0.
\end{aligned}$$

Let the dual variables corresponding to these four inequalities be

$$\alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0.$$

The corresponding dual problem is

$$\begin{aligned} \text{Minimize} \quad & 6\alpha_1 - 6\alpha_2 + 10\alpha_3 + 2\alpha_4, \\ \text{subject to} \quad & \alpha_1 - \alpha_2 + 2\alpha_3 + \alpha_4 \geq 3, \\ & -\alpha_1 + \alpha_2 - 2\alpha_3 - \alpha_4 \geq -3, \\ & \alpha_1 - \alpha_2 - \alpha_3 - 4\alpha_4 \geq -2, \\ & \alpha_1 - \alpha_2 - 3\alpha_3 + \alpha_4 \geq -1, \\ \text{and} \quad & \alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0. \end{aligned}$$

Now define

$$y_1 = \alpha_1 - \alpha_2, \qquad y_2 = \alpha_3, \qquad y_3 = -\alpha_4.$$

Then

$$y_1 \text{ is unrestricted in sign,}$$

because it is the difference of two nonnegative variables. Also,

$$y_2 \geq 0,$$

and

$$y_3 \leq 0.$$

The objective function becomes

$$\begin{aligned} 6\alpha_1 - 6\alpha_2 + 10\alpha_3 + 2\alpha_4 &= 6(\alpha_1 - \alpha_2) + 10\alpha_3 - 2(-\alpha_4) \\ &= 6y_1 + 10y_2 - 2y_3. \end{aligned}$$

The first two dual constraints are

$$\alpha_1 - \alpha_2 + 2\alpha_3 + \alpha_4 \geq 3,$$

and

$$-\alpha_1 + \alpha_2 - 2\alpha_3 - \alpha_4 \geq -3.$$

The second inequality can be rewritten as

$$\alpha_1 - \alpha_2 + 2\alpha_3 + \alpha_4 \leq 3.$$

Therefore, together they imply

$$\alpha_1 - \alpha_2 + 2\alpha_3 + \alpha_4 = 3.$$

Using
$$y_1 = \alpha_1 - \alpha_2, \qquad y_2 = \alpha_3, \qquad y_3 = -\alpha_4,$$
we obtain
$$y_1 + 2y_2 - y_3 = 3.$$
The third dual constraint is
$$\alpha_1 - \alpha_2 - \alpha_3 - 4\alpha_4 \geq -2.$$
Substituting the definitions of y_1, y_2, y_3 gives
$$y_1 - y_2 + 4y_3 \geq -2.$$
The fourth dual constraint is
$$\alpha_1 - \alpha_2 - 3\alpha_3 + \alpha_4 \geq -1.$$
Substituting the definitions gives
$$y_1 - 3y_2 - y_3 \geq -1.$$
Multiplying both sides by -1, this is equivalent to
$$-y_1 + 3y_2 + y_3 \leq 1.$$
Therefore, the standard-form dual is equivalent to
$$\begin{aligned} \text{Minimize} \quad & W = 6y_1 + 10y_2 - 2y_3, \\ \text{subject to} \quad & y_1 + 2y_2 - y_3 = 3, \\ & y_1 - y_2 + 4y_3 \geq -2, \\ & -y_1 + 3y_2 + y_3 \leq 1, \\ \text{and} \quad & y_1 \text{ is unrestricted in sign}, \\ & y_2 \geq 0, \\ & y_3 \leq 0. \end{aligned}$$
This is exactly the same dual problem obtained in part (a).

(c)

We verify that
$$\bar{x} = \left(\frac{26}{5}, \frac{4}{5}, 0\right)$$

is feasible for the primal problem.

For the first constraint,

$$\frac{26}{5} + \frac{4}{5} - 0 = \frac{30}{5} = 6.$$

For the second constraint,

$$2 \cdot \frac{26}{5} - \frac{4}{5} + 3 \cdot 0 = \frac{52}{5} - \frac{4}{5} = \frac{48}{5} \leq 10.$$

For the third constraint,

$$-\frac{26}{5} + 4 \cdot \frac{4}{5} + 0 = -\frac{26}{5} + \frac{16}{5} = -2.$$

Thus,

$$-x_1 + 4x_2 + x_3 = -2 \geq -2.$$

Also,

$$x_2 = \frac{4}{5} \geq 0, \qquad x_3 = 0 \leq 0,$$

and x_1 is unrestricted in sign. Hence $\bar{x}$ is primal feasible.

The primal objective value is

$$\begin{aligned} Z(\bar{x}) &= 3 \cdot \frac{26}{5} - 2 \cdot \frac{4}{5} + 0 \\ &= \frac{78}{5} - \frac{8}{5} \\ &= \frac{70}{5} \\ &= 14. \end{aligned}$$

Now verify that

$$\bar{y} = (2, 0, -1)$$

is feasible for the dual problem.

The sign restrictions are satisfied:

$$y_1 = 2 \text{ is unrestricted in sign},$$

$$y_2 = 0 \geq 0,$$

and

$$y_3 = -1 \leq 0.$$

Check the dual constraints:

$$y_1 + 2y_2 - y_3 = 2 + 2(0) - (-1) = 3.$$

So the first dual constraint holds with equality.

For the second dual constraint,

$$y_1 - y_2 + 4y_3 = 2 - 0 + 4(-1) = -2 \geq -2.$$

For the third dual constraint,

$$-y_1 + 3y_2 + y_3 = -2 + 0 - 1 = -3 \leq 1.$$

Therefore, $\bar{y}$ is dual feasible.

The dual objective value is

$$\begin{aligned} W(\bar{y}) &= 6(2) + 10(0) - 2(-1) \\ &= 12 + 2 \\ &= 14. \end{aligned}$$

Since $\bar{x}$ is primal feasible and $\bar{y}$ is dual feasible, weak duality gives

$$Z(\bar{x}) \leq W(\bar{y}).$$

But here

$$Z(\bar{x}) = W(\bar{y}) = 14.$$

Therefore, both $\bar{x}$ and $\bar{y}$ are optimal solutions.

Hence,

$$\bar{x} = \left(\frac{26}{5}, \frac{4}{5}, 0\right)$$

is an optimal solution of the primal problem, and

$$\bar{y} = (2, 0, -1)$$

is an optimal solution of the dual problem.

The optimal value is

$$Z^* = W^* = 14.$$